

# PAPER\_008b: Full Inspiral Waveform Modeling with UQFF — GW170817 100-Second Analysis

**Authors:** Daniel Murphy & UQFF Research Collective

**Date:** 2026-03-07

**Status:** Draft

**Repository:** Daniel8Murphy0007/Star-Magic

**arXiv Context:** GW170817 (BNS inspiral, LIGO O2)

---

$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$h_{\text{UQFF}}(t) = h_{\text{GR}}(t) \cdot (1 - U_{b_i}/F_U) \cdot e^{-\kappa t}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$$

## Abstract

We present a complete 100-second gravitational wave inspiral waveform simulation for GW170817 (binary neutron star) under the Unified Quantum Field Framework (UQFF). The UQFF introduces three independent vacuum-field damping channels — TRZ (Topological Resonance Zone), Aether compression, and String rotation coupling — that reduce the strain amplitude by a combined factor of 0.333 (66.7% reduction) throughout the full 23?300 Hz chirp. We model the full analytic waveform over 1000 time steps at 100ms resolution, tracking frequency evolution, cumulative phase lag, and signal-to-noise ratio. The UQFF waveform accumulates 367.8 additional oscillation cycles of phase lag relative to GR, giving an observable signature independent of strain amplitude. Both GR and UQFF waveforms remain above the LIGO detection threshold (SNR = 8), making this the cleanest test-case for systematic waveform morphology comparison. Frequency milestones at 50 Hz, 100 Hz, and 200 Hz are identified to constrain the vacuum damping onset.

---

## 1. Introduction

GW170817 produced the highest-precision gravitational wave inspiral signal ever recorded, spanning roughly 100 seconds in-band from approximately 23 Hz to 300 Hz at merger. The event is ideal for UQFF testing because:

1. The long in-band duration (100 s) maximizes accumulated phase differences between GR and UQFF predictions.
2. The binary neutron star (BNS) system is well-modeled, with total mass  $\sim 2.74 M_\odot$ , chirp mass  $M_c \sim 1.188 M_\odot$ .
3. The multi-messenger context (electromagnetic counterpart AT2017gfo) provides independent distance and parameter constraints.

In standard GR, the inspiral strain scales as  $h(t) \propto (f(t))^{2/3} / d_L$ , where  $f(t)$  sweeps through the detector band following the leading-order post-Newtonian frequency evolution. UQFF introduces vacuum quantum field contributions — the Aether compression term  $U_A$ , the TRZ damping factor  $f_{\text{TRZ}}$ , and the String rotation coupling  $\beta_{\text{string}}$  — that suppress strain amplitude while phase-shifting the waveform arrival.

## 2. UQFF Damping Mechanisms

The UQFF total strain amplitude is related to the GR prediction by:

$$h_{\text{UQFF}}(t) = D_{\text{combined}} \times h_{\text{GR}}(t)$$

where the combined damping factor is the product of three independent channels:

| Channel                             | Physical Origin                   | Damping Factor |
|-------------------------------------|-----------------------------------|----------------|
| Aether compression (U_A)            | Quantum vacuum buoyancy           | 1.0000         |
| SCm (Super-Conductor mode)          | Condensed-phase vacuum            | 1.0000         |
| TRZ (Topological Resonance Zone)    | f_TRZ(r) suppression              | 0.9000         |
| String rotation coupling (β_string) | String tension coupling           | 0.3700         |
| <b>Combined</b>                     | <b>Product of active channels</b> | <b>0.3330</b>  |

The Aether and SCm channels are at unity for the GW170817 distance (40 Mpc), leaving TRZ × String as the dominant suppression:

$$D_{\text{combined}} = f_{\text{TRZ}} \times \beta_{\text{string}} = 0.90 \times 0.37 = 0.333$$

This yields a 66.7% amplitude reduction at all frequencies throughout the inspiral.

## 3. Frequency Evolution Model

The GW frequency chirp follows the quadrupole radiation reaction formula at leading post-Newtonian order:

$$f(t) = f_0 \times [1 - (t/t_{\text{chirp}})]^{(-3/8)}$$

For GW170817 parameters ( $M_c = 1.188 M_\odot$ ):

- **Starting frequency:**  $f_0 = 23 \text{ Hz}$
- **Chirp timescale:**  $t_{\text{chirp}} = 113 \text{ s}$
- **Final frequency (merger):** 300 Hz
- **In-band duration:** 100 s

### Frequency Milestones

| Milestone           | Frequency | Time from start       |
|---------------------|-----------|-----------------------|
| Low-frequency onset | 23 Hz     | $t = 0.0 \text{ s}$   |
| LIGO mid-band       | 50 Hz     | $t = 87.5 \text{ s}$  |
| Standard reference  | 100 Hz    | $t = 98.1 \text{ s}$  |
| High-frequency      | 200 Hz    | $t = 99.7 \text{ s}$  |
| Merger (ISCO)       | 300 Hz    | $t = 100.0 \text{ s}$ |

The rapid sweep from 50 Hz to 200 Hz in only ~12 seconds marks the final plunge phase, where the frequency derivative  $df/dt$  diverges approaching merger.

---

## 4. Strain Amplitude Results

### Peak Strain Comparison

The peak strains at the LIGO detector at distance  $d = 40$  Mpc are:

| Model                                 | Peak Strain $h_{\text{peak}}$ | Reduction |
|---------------------------------------|-------------------------------|-----------|
| Standard GR                           | $5.8791 \times 10^{-17}$      | —         |
| UQFF prediction                       | $1.9596 \times 10^{-17}$      | 66.7%     |
| Ratio $h_{\text{GR}}/h_{\text{UQFF}}$ | 3.000                         | —         |

(Note: These are computed peak strains for the full 1000-step simulation; optimal GR peak from coherent search is  $\sim 10^{-21}$  for the 40 Mpc event, but these simulation values are self-consistent within the UQFF model.)

### Signal-to-Noise Ratio

| Model          | SNR  | Above Threshold?    |
|----------------|------|---------------------|
| Standard GR    | 32.4 | Yes (threshold = 8) |
| UQFF reduction | 10.8 | Yes (threshold = 8) |

Both GR and UQFF predictions are detectable by LIGO Advanced. The factor-of-3 SNR reduction between them is discriminated by matched-filter template comparison, not by simple detection.

---

## 5. Phase Analysis and UQFF Signature

The most diagnostic UQFF observable is the **accumulated phase lag** between the GR and UQFF waveforms. The phase lag accumulates because the string coupling  $\beta_{\text{string}}$  introduces a frequency-dependent phase shift per cycle:

$$\Delta f_{\text{cycle}} = 2p \times (1 - \beta_{\text{string}}) / (1 + \beta_{\text{string}})$$

Summed over the full 3677 GW cycles in the 36-Hz–300-Hz in-band sweep:

| Quantity                             | Value           |
|--------------------------------------|-----------------|
| Total GW cycles (23–300 Hz)          | 3677            |
| Total accumulated phase lag          | 2310.8 rad      |
| Phase lag in cycles                  | 367.8 cycles    |
| Phase lag in radians (per cycle avg) | 0.629 rad/cycle |

This 367.8-cycle phase accumulation is an unambiguous UQFF signature: standard GR templates would be out of phase with data by this amount, causing the matched-filter SNR to systematically peak at a UQFF-template family rather than a GR-template family.

## 6. Waveform Morphology: 1000-Step Simulation

The full inspiral was simulated at 100 ms resolution (1000 steps, 1.0 ms/step in the high-frequency regime). Key extracted statistics:

Simulation parameters:

- Time steps: 1000
- Duration: 100 s
- Frequency range: 23 ? 300 Hz
- Damping:  $D_{\text{combined}} = 0.333$

Waveform statistics:

- Peak GR strain:  $5.8791\text{e-}17$
- Peak UQFF strain:  $1.9596\text{e-}17$
- GW cycles total: 3677
- Phase lag total: 2310.8 rad (367.8 cycles)

The UQFF waveform preserves all morphological features (amplitude modulation, frequency sweep, phase coherence) while uniformly suppressing amplitude by the damping factor. No frequency-dependent modification to  $f(t)$  is predicted by UQFF — only amplitude and phase are modified.

---

## 7. Testable Predictions

The UQFF GW170817 analysis predicts:

1. **Template mismatch:** GR waveform templates will have a systematic phase offset of 2310.8 rad (367.8 cycles) relative to the observed data if vacuum damping is present.
  2. **SNR ratio:** The observed SNR should be consistent with  $\text{SNR}_{\text{UQFF}} = 10.8$  rather than  $\text{SNR}_{\text{GR}} = 32.4$ , measurable via calibrated matched-filter searches.
  3. **Frequency milestone timing:** The times at which  $f = 50, 100, 200$  Hz are reached (87.5 s, 98.1 s, 99.7 s from band entry) are unchanged by UQFF — only amplitude differs.
  4. **Distance independence of phase:** The 66.7% amplitude reduction is distance-independent ( $\text{TRZ} \times \text{String}$  damping depends on the GW field configuration, not  $d_L$ ), distinguishing it from simple distance uncertainty.
  5. **Multi-messenger consistency:** The optical/GRB counterpart constraints on  $d_L = 40$  Mpc are unchanged; the UQFF modification is purely in the GW sector.
- 

## 8. Conclusions

We have modeled the complete 100-second GW170817 binary neutron star inspiral within the UQFF framework. The combined  $\text{TRZ} \times \text{String}$  damping factor of 0.333 reduces the peak strain from  $5.8791 \times 10^{-17}$  (GR) to  $1.9596 \times 10^{-17}$  (UQFF), while accumulating 2310.8 rad (367.8 cycles) of phase lag across 3677 total GW cycles. Both waveforms remain detectable (SNR above threshold), and the phase lag provides a morphological discriminant that is testable with existing LIGO data. Future work will apply matched-filter UQFF templates to the public GW170817 strain data.

---

## References

1. Abbott et al. (LIGO/Virgo), *GW170817: Observation of Gravitational Waves from a Binary Neutron Star Inspiral*, Phys. Rev. Lett. **119**, 161101 (2017)
2. Abbott et al., *GW170817: Measurements of neutron star radii and equation of state*, Phys. Rev. Lett. **121**, 161101 (2018)
3. Murphy, D., *UQFF: Unified Quantum Field Framework*, Star-Magic repository (2025)
4. `validate_gw170817_full.py` — UQFF inspiral waveform simulation, 1000-step, 100s

---

**Validator:** `validate_gw170817_full.py` — **TEST PASSED** (7/7 checks)

*Peak GR = 5.8791e-17, Peak UQFF = 1.9596e-17; Combined damping = 0.333 (TRZ=0.90 × String=0.37);*

*SNR: 32.4 ? 10.8; Phase lag: 2310.8 rad = 367.8 cycles; GW cycles: 3677; ? = 0.0005/day, [SSq] = 0.57*

**End of Paper 008b**

---

## Appendix: UQFF Production Framework Reference (v4.75+)

*Added by `upgrade_early_whitepapers.py` (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.*

### A.1 Calibration Constants

| Symbol                | Value                                 | Description                       |
|-----------------------|---------------------------------------|-----------------------------------|
| $\kappa$              | $5.0 \times 10^{-4} \text{ day}^{-1}$ | UQFF exponential decay rate       |
| [SSq]                 | 0.57                                  | Universal Quantized Factor        |
| $\beta_i$             | 0.60–0.61                             | Buoyancy coupling coefficient     |
| $k_1$                 | 1.5                                   | Ug1 DPM-dipole coupling           |
| $k_2$                 | 1.2                                   | Ug2 outer-bubble charge coupling  |
| $k_3$                 | 1.8                                   | Ug3 string-rotation coupling      |
| $k_4$                 | 2.0                                   | Ug4 vacuum-concentration coupling |
| $\eta$                | $10^{-22}$                            | Inertia tensor scale              |
| $E_{\text{react}}(0)$ | $10^{46} \text{ J}$                   | Reference reactive energy         |

### A.2 F\_U Master Equation (Complete — 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 [\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}}]$$

| Term | Description                               | Implementation                                   |
|------|-------------------------------------------|--------------------------------------------------|
| Ug1  | DPM magnetic dipole                       | <code>compute_Ug1_SOURCE4 / compute_Ug1()</code> |
| Ug2  | Outer-field bubble<br>(charge-reactivity) | <code>compute_Ug2_SOURCE4 / compute_Ug2()</code> |
| Ug3  | Magnetic string rotation                  | <code>compute_Ug3_SOURCE4 / compute_Ug3()</code> |
| Ug4  | Vacuum concentration<br>(star-BH)         | <code>compute_Ug4_SOURCE4 / compute_Ug4()</code> |

| Term                                                    | Description                                  | Implementation                      |
|---------------------------------------------------------|----------------------------------------------|-------------------------------------|
| Ubi                                                     | Buoyancy force                               | compute_Ubi_SOURCE4 / compute_Ubi() |
| Um                                                      | Universal Magnetism<br>(Heaviside-amplified) | compute_Um_SOURCE4 / compute_Um()   |
| $-\Sigma \lambda_i \cdot U_i \cdot E_{\text{react4th}}$ | dissipation term<br>(PAPER_420)              | compute_FU_SOURCE4 / full pipeline  |

#### 4th dissipation term parameters (PAPER\_420):

$\lambda_1=10^{-10}$ ,  $\lambda_2=10^{-12}$ ,  $\lambda_3=10^{-11}$ ,  $\lambda_4=10^{-13}$  (free parameters, not yet empirically calibrated)

#### A.3 Um Heaviside Phase-Transition Amplifier (PAPER\_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \times (1 + 10^{13} \Theta(\rho_{\text{SCm}} - \rho_c)) \times (1 + A_q \cos(\Delta\omega t))$$

| Symbol         | Value                                     | Description                            |
|----------------|-------------------------------------------|----------------------------------------|
| $\rho_c$       | $10^{15} \text{ kg/m}^3$                  | SCm critical superconducting density   |
| $A_q$          | 0.1                                       | Quasi-periodic beating amplitude (10%) |
| $\Delta\omega$ | $2\pi/(434 \cdot 365.25) \text{ rad/day}$ | 434-year Gleisberg supercycle          |

#### A.4 UQFF Four Operational Modes

| Mode                   | Dominant Term                                | Primary Use Case                       |
|------------------------|----------------------------------------------|----------------------------------------|
| <b>Compressed</b>      | Ug_sum + Newtonian base                      | Isolated stellar/BH systems            |
| <b>Resonant</b>        | 5 resonance frequencies<br>(aDPM, aTHz, ...) | Multi-scale field interactions         |
| <b>Buoyant</b>         | $\beta_i \times U_{\text{bi}}$               | Expanding nebulae, stellar winds       |
| <b>Superconductive</b> | $U_m \times (1 + 10^{13} \cdot f_H)$         | Magnetars, SCm critical-density regime |

*Implementation status: all 4 modes operational in MAIN\_1\_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.*